

13 Descent versus Extinction: The Reception of Darwinism in Estonia

Ken Kalling and Erki Tammiksaar

To describe the reception of Darwinism in what is now the Republic of Estonia calls for a flexible approach in order to uncover the different layers emerging from the complex development of local society at the end of the nineteenth and the beginning of the twentieth century. In 1859, when Charles Darwin's *Origin of Species* appeared, there were three distinguishable players in the local political theatre: the Russian state, the Baltic German elite, and the subjugated Estonian people. This situation lasted until 1920, when Estonia gained de facto independence. Until then the Russian Baltic Provinces (Estland, Livland and Kurland, the northern part of which is modern Estonia and the southern part Latvia) had, from the beginning of the eighteenth century, been controlled by the Russian central government. The Russian state in the period discussed in this article was gradually implementing a Russification policy whose objective was to erode both the dominant position of the Lutheran Church and the autonomy of the local non-Russian community.

The so-called Baltic Germans, who constituted the upper layers of society, were, as a group, the second player on the local political field. The Germans had entered the stage in the thirteenth century, during the so-called Baltic Crusade. Although few in number, they were the key figures in local economic and political life, and creators of the cultural and ideological environment.

The indigenous population of Estonians, who for centuries had been dominated by German-speaking landlords, was only starting to emerge as a modern European nation in the middle of the nineteenth century. This process of nation-building involved treading a path between the often contradictory interests of the two dominant players, the local Germans and the Russian state. Despite growing Russian pressure on Baltic autonomy, Germans and Estonians failed to cooperate effectively because of mutual mistrust. Many Estonians, too, mistrusted the Lutheran Church, which they viewed as a 'German' phenomenon, a mechanism of feudal oppression. When in the 1880s, Russification began to affect indigenous Estonians, it became increasingly clear that the Russian authorities were not just allies against the Germans but also a threat to national emancipation.

The constantly increasing dominance of Russia was relieved only by the 1905 Russian Revolution, which authorized the use of local languages, and the Russian Revolution and Civil War (1917–20), which brought Estonia independence. As a result the landed wealth of the Baltic Germans was confiscated and the country freed from feudalism.

Thus, the reception of Darwinism in Estonia must be traced by studying the two local cultural groups, Estonians and Germans, while understanding the perceived threat of Russian dominance, which influenced each local culture in different ways. The Estonian and German receptions of Darwinism were different due to the different economic, social and political environments that these two socio-ethnic groups occupied. Estonians were predominantly peasants, with a modest elite that had only recently developed a scientific vocabulary and a literature. Indeed, Darwin's *Origin of Species* has yet to be translated into Estonian!

The intellectual and ideological centre of Baltic-German identity was the Lutheran University of Dorpat (now Tartu), where German was the language of instruction. This university was home to numerous outstanding scholars, some of whom, like Karl Ernst von Baer, had distinguished themselves in pre-Darwinian evolutionary science.

Both the Baltic-German and Estonian receptions of Darwin were politically inflected and defensive. This can be explained by acute political tensions in the Russian Baltic Provinces during the period discussed. Of the two, Estonian political thought seems to have been the more biologized, inasmuch as it looked to biology to justify national emancipation, and it was also concerned with demographic issues, i.e. self-perception as a 'small nation', foreshadowing Estonian social discourse for decades (Kalling 2007b).

The existing literature on the Baltic-German and Estonian receptions of Darwinism has focused on several local scholars. One should mention, first, Georg von Seidlitz (1840–1917), one of Darwin's advocates in the Russian Empire, the paleontologist Alexander von Keyserling (1815–91) and the embryologist Karl Ernst von Baer (1792–1876). Seidlitz's views on Darwin have been analysed by Leonid Bliakher (1971) and Maie Valt (1975). Baer's contribution on Darwinism and his criticism of it have been studied by many scholars, in most detail by Remigius Stölzle (1897), Boris Raikov (1968), M. Valt (1977) and L. Bliakher (1982). A. von Keyserling's views on Darwin have been analysed by B. Raikov (1959). Estonian themes appear in Alexander Vucinich's study of the reception of Darwinism in Russia (1988). The Estonian reception of evolutionary theory has been touched upon modestly, mainly in Soviet literature in connection with the spreading of materialist ideas (Laul 1956; Raid 1978), and by one of the authors of this chapter, concentrating on the reception of Darwinism by Estonian elites (Kalling 2007a).

These works are mainly empirical history of science or focused on issues of theoretical biology. The present study aims to break new ground and concentrate on the social and political reception of evolutionary theory in the region. The first part of the article is dedicated to the German-language reception of Darwinism by Baltic scholars in the second half of the nineteenth and beginning of the twentieth century. The second part deals with the spread of the new ideas among the Estonian elites and their reflection on national ideologies, including that of the pre-war Estonian Republic. We do not here deal with the post-World War II Soviet era, when the situation in Estonia paralleled general trends in the USSR, in particular what might be called a von Baer 'cult' in the 1970s when the 100th anniversary of the famous scholar's death was celebrated (Tammiksaar 1999a).

The Baltic-German Reception

Acceptance of the theory

The reception of the *Origin of Species* in Russia coincided with the beginnings of the so-called Panslavistic movement, a nationalist phenomenon that gradually gained strength in the political reasoning of the Empire. One of the aims of the movement was to weaken and finally put an end to the influence of 'Germans' (non-Russians generally) in the state. Up to the 1860s the German-speaking upper layers in Baltic society felt themselves firmly in control. But in the second half of that decade the situation started to change. Then, the Panslavist-backed state launched a more concentrated attack on the Baltic nobility and their privileges, taking advantage of the emerging Estonian and Latvian national movements, which opposed the local feudal system.

The Germans, now under attack, reacted in pain to every attempt to diminish the use of the German language and limit the position of the Lutheran Church and, in doing so, became even more attached to their language and faith. As early as 1843, after the Russian authorities attempted to close the faculty of theology at the Lutheran University of Tartu, one of the leading Baltic scientists, K. E. von Baer, wrote:

Since I was convinced that after a century there would hardly be a Protestant church, at least no Protestant province in the [Russian] Empire if these first steps were approved [by the Tsar] I was determined to return to Germany with my family, even if I had to do it on foot. For I didn't want to accept the responsibility for my descendants having to convert to the Eastern Church.¹

This situation and the attitudes it bred explains why theologians had relatively more influence at the University of Tartu than in other German universities, a situation which had repercussions in the reception of Darwinism among local educated circles and even less educated ones. An example of this negative attitude towards the new scientific theory can be illustrated by the so-called 'Schleiden case'.

World-famous as one of the founders of cell theory, Matthias Jakob Schleiden (1804–81) was elected professor of plant physiology and anthropology at the University of Tartu in 1863. Already famous in Germany for his fight to free the natural sciences from the influences of religion, he was the first to propagate Darwinian ideas in the Baltic Provinces. Schleiden's lectures at the University of Tartu on materialism, containing aspects of the evolutionary theory, were enormously popular (800 people attended them) (Jahn and Schmidt 2005, 166–67). His materialist rhetoric and his critiques of religion drew Schleiden

¹ 'Ueberzeugt, dass wenn diese Einleitungen sanctionnirt würden, in einem Jahrhundert kaum noch eine protestantische Kirche, wenigstens keine protestantische Provinz im Reich sein würde, war ich entschlossen mit meiner Familie nach Deutschland zurückzukehren, und wenn es auch zu Fuss sein müsste, denn die Verantwortung wollte ich nicht auf mich laden, dass meine Nachkommen zur morgenländische Kirche übergehen müssten' (K. E. von Baer to Alexander Theodor von Middendorff, 30 August 1843, Hessisches Staatsarchiv Marburg, *Estländische Ritterschaften papers*, folder 702, no. 321).

into strong opposition to local theologians, but also into conflict with naturalists tied to vitalism. They viewed him as both an atheist and a political threat – a representative of Tsarist rule (Bidder 1959, 173–79). Thus was Schleiden's fate decided and, although backed by the Russian Court, he left Tartu in 1864.

The antimaterialist position of the elites influenced the local popular reception of science, for instance in local newspapers, where there can be found only a sample of the debate between the supporters and opponents of Darwinism, mainly overviews from the German press (Wagner 1873; Baer 1873; 'Wigand' 1874; Vogt 1875). The modest representation of a much discussed issue can perhaps be explained by the bias of the editors. The only scholars of Baltic-German origin to deal with the issues of Darwinism in a popular way – K. E. von Baer and G. von Seidlitz – published their works in German newspapers (Baer 1873; Seidlitz 1874a; b). The characteristic stance of the newspapers can be witnessed in the proceedings (*Sitzungsberichte der Naturforscher-Gesellschaft bei der Universität Dorpat*) of the amateurish Tartu Naturalists' Society,² which never touched on evolution (at least during the years 1870 to 1900), nor was the issue raised at meetings of the society, nor did it pay any attention to Darwin's death.

Such public neglect cannot be generalized to the whole Baltic community because Darwinism was certainly present in more scholarly discussions of evolutionary theory. For example, K. E. von Baer was in a privileged position, having learned about the soon to be published new theory during his visit to London in the autumn of 1859 (Burkhardt and Smith 1993, 311–12). Local scientists learned about the new theory mainly from German-language sources (e.g. *Das Ausland* (Foreign lands)) and debated its merits in private correspondence (Tauben von der Issen 1902).

The environment where evolutionary theory enriched local discussions, however, was the local specialized weekly and monthly journals. Carl Johann von Seidlitz (1798–1885), a Livonian landlord and medical doctor, published a long overview of Darwin's ideas on animal husbandry in an agricultural weekly, *Baltische Wochenschrift* (The Baltic weekly) (Seidlitz 1868). The same journal also printed his overview of Darwin's study on the *Descent of Man* (Seidlitz 1871). Seidlitz stood on the side of Darwin. Among the few Baltic-German scholars to share his views (and accordingly Darwin's) was a philologist named Arthur Amelung (1840–74), who wrote an article in a liberal weekly, *Baltische Monatsschrift* (The Baltic monthly) on the possible applications of Darwin's theory in language studies (Amelung 1871). This work could be viewed as one of the few creative approaches to evolutionary theory present in the local realm. Amelung suggests that the emergence and evolution of species could be compared with that of languages. He is still not sure if languages might have a common ancestor. Moreover, he was convinced that mechanical parallels cannot be drawn between the methods of biology and linguistics, inasmuch as developments within the latter are strongly influenced by non-natural phenomena, i.e. the social and cultural environment.

² 'Naturforscher Gesellschaft'

The most famous propagator of Darwinism in the Baltic Provinces was the zoologist, Georg von Seidlitz, son of Carl Johann von Seidlitz, and an ardent Darwinist. This can be appreciated from the fact that he named his first son (born in 1871) Gerhard Karl Lamarck Darwin (Stackelberg 1929–30, 255). Georg von Seidlitz was the first to lecture on Darwin's evolutionary theory in Russia from 1870 to 1877. He also published a German-language university textbook on the subject (*Die Darwin'sche Theorie: Elf Vorlesungen über die Entstehung der Thiere und Pflanzen durch Naturzüchtung* (Darwin's Theory: eleven lectures on the descent of animals and plants through natural selection), 1871, 2nd edn 1875), and offered a series of lectures on evolutionary theory meant for a broader audience, which was published in the *Baltische Monatsschrift* (1871b). In his lectures, he limited his attention to teaching and explaining natural sciences through the methods of Darwin, and avoided anticlerical or materialist statements that lent themselves to political debate. He made this point himself: 'These are academic lectures of the first semester 1870 delivered to students of all faculties of the university, and they aim to be nothing more than their title announces.'³ In view of the Baltic social environment he came from (and perhaps taking Schleiden's fate into account), his caution can be understood. Still, he directed a discreet message to his compatriots to the effect that they should forget about classifying themselves among civilized nations if they did not accept Darwin's theory: 'Nowadays the cultural level of a whole nation as well as the degree of education of the single individual is recognized by their degree of achievement in the explanatory natural sciences.'⁴

An unusual perspective on the reception of Darwin's theory in social sciences is offered by the case of Paul von Lilienfeld, governor of Kurland Province (now Latvia). Although he had no university degree, he was a prolific writer and compiled a monograph on the social science of the future, published anonymously in 1873 (Lilienfeld 1873). He was fascinated by developments in the natural sciences and tried to show that the processes of human society (*socialer Organismus*, literally 'social organism') copy the laws of physics, chemistry and biology. The hierarchical stratification of races, for example, can be explained as stages in the development of the nervous system. Although he denied it, Lilienfeld borrowed ideas from Darwin, Haeckel and other evolutionary writers. His writing was controversial and was ridiculed by Alexander von Oettingen (1827–1905), professor of theology at the University of Tartu (Oettingen 1873; 1874).

Karl Ernst von Baer's reception of Darwinism

Von Baer was the most prominent natural scientist to emerge from Estonia. His attitude towards Darwin's teachings was well known because his embryological work had been an authority for Darwin himself, in particular von Baer's version

³ 'Es sind akademische Vorlesungen aus dem I. Semester 1870, die vor Studirenden aller Facultäten gehalten wurden, und nichts anderes zu sein beanspruchen, als was ihr Name besagt' (Seidlitz 1871a, vi).

⁴ 'Heut' zu Tage erkennt man sowohl die Culturstufe eines ganzen Volkes als auch den Bildungsgrad des einzelnen Individuums an der Stellung, die sie zu den Errungenschaften der erklärenden Naturerkenntniss einnehmen' (Seidlitz 1871b, 245).

of recapitulation theory. Von Baer was a crucial figure in the Baltic-German community and his ideas on Darwinism were controversial. Baltic-German Darwinists were reluctant to accept their idol's anti-Darwinism. For example, Georg von Seidlitz, in the first instalment of his lectures (1871a, 186–88), filled a whole chapter explaining why von Baer could not be viewed as an opponent of Darwin's theory.⁵ This was the same period in which several German naturalists tried to prove that von Baer could not be viewed as a *supporter* of Darwinism (e.g. Wagner 1873).

It has to be admitted that von Baer's statements on Darwin were contradictory. This can be explained by the tentative nature of his position: he had not made a thorough study of the topic, although it is evident from the correspondence between Darwin and Huxley that von Baer was 'much interested' in the works of the British naturalist (Burkhardt and Smith 1993, 311–12). The problem for von Baer was that because of the lack of experimental proof he preferred to regard evolutionary theory as still just a hypothesis. Such an attitude can be seen in all von Baer's comments on Darwin's ideas. It is also true that he became gradually more and more sceptical of Darwin's conclusions. This can be seen in his correspondence with physiologist Friedrich Bidder⁶ and zoologist Moritz Wagner, where he refers to convinced Darwinians such as Georg von Seidlitz and Ernst Haeckel as 'Ultra-Darwinists'.⁷

Here one may ask why von Baer joined the critics of Darwin? Was it just insufficient empirical proof or was there some other important motive? It has been written that von Baer became critical of Darwin in response to the strong statements in support of the evolutionary theory made by Haeckel in 1866. It was, in fact, Haeckel's *Generelle Morphologie der Organismen* (General morphology of organisms) which worried von Baer, because Haeckel's approach to biogenetic law contradicted von Baer's own recapitulation theory in many aspects (Valt 1975, 21).

On the other hand, it is evident that von Baer's approach had already been sceptical earlier, and that was due to reasons deriving both from his teleological approach to explaining nature and also political issues emerging from the situation in the Russian Baltic Provinces. There were several scientific arguments that von Baer used against Darwin's theory. Although he did not deny transformation, he was sure that no visible changes of species had taken place during the last 2,000 years. For von Baer, the paleontological evidence from different geological layers also seemed to be too scattered to establish relationships among, or common ancestry of, living creatures. He was also convinced that a species cannot change externally inasmuch as its development has been settled embryologically (Baer 1876).

⁵ Oddly enough, this 'patriotic' approach surfaced much later – in Soviet Estonia – where von Baer had also been elevated to a position of local hero in the history of science (Valt 1977).

⁶ K. E. von Baer to Friedrich Bidder [n.d.], Hessisches Staatsarchiv Marburg, *Estländische Ritterschaften* papers, folder 702, no. 321.

⁷ K. E. von Baer to Moritz Wagner, 19 June/1 July 1868, Universitätsbibliothek Giessen, Handschriftenabteilung, von Baer folder.

Von Baer represented a society that valued hierarchical privilege and the autonomy of its members. He had for years worked for the interests of the Baltic-German community, especially when residing in St Petersburg (1834–67), the capital of the Empire (Tammiksaar 1999a). The use of Darwin's theory to explain the functioning of human societies, evidently eroding the moral grounds of the present way of life, caused ambivalent feelings in von Baer. In 1866 he described them to a friend: 'Darwin, this mammoth of modern times, stirs up everything and confuses the head of an old conservative. I believe the world must soon collapse.'⁸ As the years passed, von Baer's conviction grew that humankind was deteriorating due to the introduction of Darwinism into social discourse. Thus he felt it was his duty to intervene. He tried to defeat evolutionary theory using scientific methods. His work on this topic was finished in 1875 and published in 1876 – the year of his death – under the title *Ueber Darwin's Lehre* (On Darwin's doctrine) (Baer 1876). Baer criticized Darwin's views, using empirical scientific material, but keeping in mind also the social aspects of Darwinism. In 1876 he wrote to Johannes Huber, also a critic of Darwin, explaining his motives:

Indeed the future also seems dark to me, when I hear how nowadays people often scoff jubilantly at all religion, all moral law and all values generally as outdated superstition. It is exactly this aspect that makes Darwinism dangerous. Apart from that one may believe in transmutations as much as one likes, for there are no proofs either for or against them. For we can only recognize that what happened a number of millennia ago was different from now. But I must doubt very decidedly that humankind will be happier when it abandons all values. To be sure, the human being is the only being that has an understanding of values. If you rob human beings of values then you make them into animals. I touched upon this problem only slightly in my writing, because I wanted to speak only as a naturalist. Had I done the opposite, the Darwinians would have immediately cast me aside.⁹

In this way, von Baer acted in congruence with his social surroundings. He opposed Darwin because he concluded that values and aspirations threatened by

⁸ 'Darwin, dieses Mammut der Neuzeit, bricht überall durch und macht einem alten Konservativen den Kopf wüst. Ich glaube, die Welt muss bald untergehen' (Lukina 1975, 204).

⁹ 'Aber freilich erscheint auch mir die Zukunft dunkel, wenn ich höre, wie jubelnd man jetzt häufig alle Religion, alles Sittengesetz und überhaupt alles Ideale als antiquierten Aberglauben verspottet. Dies ist es ja eben, was den Darwinismus gefährlich macht, sonst mag man an Umwandlungen glauben, wie man will, es giebt weder Beweise dafür noch dagegen. Denn was vor einer Reihe von Jahrtausenden vorgegangen ist, davon können wir nur erkennen, dass es anders war als jetzt. Dass aber das Menschengeschlecht glücklicher wird, wenn es sich von allem Idealen ablöst, muss ich sehr entscheidend bezweifeln. Der Mensch ist ja das einzige Wesen, das Sinn für das Ideale hat. Es dem Menschen zu rauben, heisst ihn zum Tiere machen. Wenn ich diese Seite in meiner Schrift kaum berührt habe, so geschah es, weil ich durchaus nur als Naturforscher sprechen wollte. Im entgegengesetzten Falle hätten mich die Darwinisten gleich beiseite geworfen' (Stölzle 1897, 674).

Darwinism were important for both the natural sciences and human society. Von Baer hoped that the harmony of knowledge and religion could be achieved.

On the occasion of von Baer's death there were speculations that he had embraced religion at his deathbed (Engelhardt 1876). This statement was rejected by Georg von Seidlitz, at the time von Baer's doctor, who stated that his patient was deaf during his last weeks and could not converse in the way theologians now claimed to show that he had turned to God (Seidlitz 1876). For von Baer, God was not a 'creator' but an 'initiator' – in a sense, God was nature itself, viewed in accordance with idealistic, teleological principles (Raikov 1968).

In conclusion we can state that social issues and attitudes prevailing in the Baltic Provinces favoured making the anti-Darwinian stance the dominant one among the Baltic Germans, including scholars. In such circumstances it is understandable that, for example, a famous ethologist born into the Baltic nobility, Jakob von Uexküll (1864–1944), asserted that Darwin's theory was wrong and not subject to proof (Uexküll 1910). Similar attitudes were found among other Baltic German scientists such as the physiologist Gustav von Bunge (1844–1920), a founder of biovitalism (Bunge 1887, 14; 1901, 11), and the chemist and Nobelist, Wilhelm Ostwald (1853–1932), known for his monistic treatises (Ostwald 1912). Darwinian teachings did not have a sympathetic hearing in the extremely conservative Baltic-German society, due to the threats that evolutionary theory seemed to offer to its social and ideological stability.

'A nation with a Natural History': The Estonian reception of Darwinism

The first glimmer of Darwin's new doctrine in the Estonian language was in connection with the short stay of M. J. Schleiden in Tartu from 1863 to 1864. The negative reception of the famous Darwinian biologist by the German-speaking elites was mirrored also in the Estonian-language press. Johann Voldemar Jannsen (1819–90), a journalist active in the national movement (sometimes viewed in Estonian historiography as a pro-German political figure), admitted to being gladdened that Schleiden could not speak Estonian, a guarantee that the 'First Son of the Old Ape'¹⁰ could not sow the seeds of the radical ideas of Darwin among the peasants (that is, Estonians) (Jannsen 1863, 388).

The opposite standpoint – the materialist and antireligious – was also present. Friedrich Reinhold Kreutzwald (1803–82), a medical doctor and the author of the national epic *Kalevipoeg* (Kalev's Son), welcomed the arrival of Schleiden, hoping that this event would bring some light to Tartu, this 'dark den'.¹¹ That was how he characterized the local university because of the domination of clerical spirit within its walls (*Fr. R. Kreutzwaldi kirjavahetus* (Correspondence of Fr. R. Kreutzwald), IV, 1959, 146). Kreutzwald could be viewed as an early radical and, indeed, materialist concerns remained central in the reception of natural sciences for the coming decades, in spite of a decline in both the Estonian national movement and the influx of sensitive scientific topics into public discourse.

¹⁰ 'Vana ahvi esimene poeg'.

¹¹ 'Pime urgas'.

In these years, there was at least one popular booklet touching on evolutionary theory, *Naturwissenschaftliche Naturbücher* (Scientific books on nature), by Aron Bernstein, based on his German original, in which he avoids taking sides with respect to Darwin (who is mentioned by name) (Grentzstein 1879–80, 68).

The new surge in the Estonian national movement started in the 1890s. Now the leaders were predominantly people with academic backgrounds, which was a favourable atmosphere in which to make evolutionary theory heard through the influence of the natural sciences on their radical and/or nationalist rhetoric. Biologization of national politics can be best traced in the ideology of the Estonian anti-alcohol movement. The latter, born at the end of the 1880s, was initially based on moral and economic concerns, but soon ideas deriving from the medical sciences began to predominate. An important line of argumentation was based on degeneration theory first, genetics and eugenics later. These were discussed in the context of a social-Darwinist understanding of national emancipation, reinforced by the so-called small nation's self-perception: there were just one million Estonians and the number showed no signs of increase. Low birth rates and small population numbers compared with the superiority of (Russian and German) neighbours caused a national inferiority complex and a number of existential fears. In the face of a perceived threat of extinction there emerged the foundations for a strong presence of eugenic and pronatalist sentiment in Estonian society.

The popular anti-alcohol movement gradually fell under the control of the national elites, who viewed it as a vehicle for organizing political work. Inasmuch as national politics was shut down by Russian authorities – especially before the 1905 Revolution – civil organizations like music and abstinence societies played an important role in nation-building. In the politicized context of the anti-alcohol movement, the programme of eugenics gained strength. In 1890, Villem Reiman (1861–1917), a pastor and activist in the abstinence movement, was using Darwin's theory as an argument for the degenerative influence of alcohol on the individual and the population (Reiman 1890). Reiman was also aware of the Darwinian current that saw altruism as one of the key mechanisms in evolution, and he suggested this approach for the Estonian community (*Eesti Kirjanduse Seltsi Aastaraamat* (Annual of the Estonian Literary Society) 1912, 26).

A paradox was that Reiman, in the final years of the nineteenth century, although relying on Darwin in his argumentation, represented sectors of the Estonian elite that were not ready for the materialistic concept (closely associated with Darwinism at that time). He, together with a number of Lutheran pastors, belonged to the so-called moderate wing in the national movement. Later – after the 1905 Russian Revolution had enabled an Estonian political life – this group became the first national liberal party in the Estonian political landscape.¹²

In Estonian historiography the moderates are often linked with the Estonian Students Society (hereafter, EÜS),¹³ established in 1870, but also with the newspaper, *Postimees* (The Courier). These institutions and the people affiliated with them are generally taken as the dominating force among Estonian political

¹² Eesti Rahvuslik Eduerakond.

¹³ Eesti Üliõpilaste Selts.

movements of that time, especially prior to the revolutionary changes of 1905. Their official slogan was 'national idealism', a denial of compromise with the German and Russian forces. This often meant keeping a low profile in broader political issues and sticking to the realm of society activities. This idealism was also reflected (at least initially, when the movement was more under the control of Lutheran pastors) in their ideology.

Soon this sector of the national elites had to rethink that part of its political credo that had negated materialism. The change took place – as had already happened in Reiman's case – due to the social pessimism reflected in the abstinence movement and the early ideas of eugenics. In such a situation there were also optimistic viewpoints offered by Darwinian explanations, which became available to the nationalists in the discussion of ethnic and social matters. It was stated that there seemed to be no 'Law of Nature' that condemned a small organism to decay; indeed, the latter could even be winners in the evolutionary process. Thus the basis of national history could be switched from the traditional, historical one to a biological one. 'Natural History', therefore, seemed to present the possibility of forgetting to be a 'nation without history' (as Estonians, particularly the peasants, were deemed to be) (Luiga 1910). 'Natural Laws' were now declared to have precedence over man-made ones. Thus Darwinism, a scientific theory, did not have to bow before the will of parliaments (Pill 1910, 58). Evolutionary theory could come to support national emancipation through the characteristic liberal ideology of meliorism (Karjahärm and Sirk 1997, 230): If apes could evolve into humans then why couldn't Estonians transform themselves from a peasant folk into a modern Western society (Koppel 1910, 3)?

As a result, the ideology of moderate to liberal nationalists (all the authors cited in the previous passage belong to this group) shifted from anti-materialism towards materialism (and evolutionary thought). The change was sudden: as late as 1902 one of their publication series (the *Sirvilauaa* or calendar issued by the EÜS) showed strong anti-materialist sentiment (Tischler 1902). But by 1907 the EÜS had created a section of natural sciences and, in 1910, a volume of the annual album of the EÜS was dedicated to the centenary of Darwin's birth (*Eesti Üliõpilaste Seltsi Album*, Koppel 1910). It had an introduction by Henrik Koppel (1863–1944) – later the first rector of the Estonian-language University of Tartu – and a concluding chapter by Mihkel Pill (1884–1951).

This ideological change was noticed by political opponents and elicited an ironic comment (hinting at the eugenics ideology accepted by several of the liberal leaders): 'Before 1905 the national wing [i.e. the moderates] denounced Marxism for its appetite for minor ideals. After 1905, when the nationalists started to construct their own ideology, it became evident that their high ideals came down to cattle breeding.'¹⁴ This must be considered in the light of another early comment by which Estonian national thought, under the strong influence

¹⁴ 'Enne 1905. a. tembeldas rahvusline vool marxismi tema aadete lameduse pärast "kõhuorjuseks". Pärast 1905, kui rahvuslus ise oma ideoloogiat hakkas selgitama, tuli ilmsiks, mis tema enda "kõrge pilvedetagune aade" oli.' See oli – karjakasvatus (Laaman 1915, 41).

of German ideas, became similarly biologized, and infected by racial theories (Kruus 1921, 89).

Haeckel replaces Darwin

German influence became even more evident as another important player in the Estonian national and political landscape obtained its ideological support from the German reception of Darwinism, in particular Ernst Haeckel. At this point it has to be stressed that evolution was quickly accepted in Estonia, having been introduced not by the EÜS, but by a radical student group later known under the name *Ühendus* (Union).¹⁵ The radicals included several students of medicine who are often portrayed as the first proponents of evolutionary theory in the Estonian language (Laul 1956; Raid 1978; Sutt 1975). It has to be admitted that the main objective of so-called radicals like Andres Alver (1869–1903) and Richard Aavakivi (1873–1906) was to introduce a materialistic message. In such a context, popular diffusion of evolutionary theory was just a means towards the materialist end. Topics such as the emergence of human civilization (Aavakivi 1907) or death as a biological phenomenon (e.g. Hellat 1910) were also discussed.

Yet the situation cannot be limited to popularization only. More complex reactions appeared as the moderate and radical nationalists debated Darwinism. In 1900, R. Aavakivi polemicized with an Estonian theologian and philosopher, Aleksander Kaelas (1880–1920), who had attacked evolution with primitive populist arguments in a family weekly, *Linda*. This journal represented the views of the moderate/liberal group (Kaelas 1900). Aavakivi's victory over his opponent was evident, as the consensus view already supported his ideas; that is, those of Darwin (Ramul 1921, 56).

It was paradoxical for the future that from the initially radical, leftist and even anti-German student movement, *Ühendus*, there later emerged several rather reactionary and influential political figures in the interwar Estonian Republic, representing monist and even (proto) fascist ideas. Of these the most noteworthy was Konstantin Päts (1874–1956), the first president of the country, who rose to this position through a *coup d'état* in 1934, and who introduced a corporative political system with eugenicist ideology and pronatalism as crucial factors supporting the antidemocratic, organicist concept of a 'national entity' (Kruus 1940).¹⁶ In this way the optimistic social-Darwinist characterization of the emergence of the state, which had gained ground in the War of Independence (1918–20) against both (Soviet) Russians and (Baltic) Germans, symbolizing the Darwinian strength of the small nation (Ploompuu 1920, 30), was devalued into a concern for low birth rates. It was feared that if eugenic and pronatalist means were not adopted, the victory in the war would be lost in the 'biological struggle' among nations.

The way from radicalism to reaction, from social-democratic to monist solidarity, emerged from a search for a so-called 'third way' in national politics, identified as a correction to the initial post-1905 pattern of Estonian politics, limited to national-liberals and social democrats. The contribution of the socialists

¹⁵ Endiste Eesti Üliõpilaste Selts 'Ühendus'.

¹⁶ 'Rahvusterviklikkus'.

to Darwin's reception in Estonia was a modest one, represented by several translations (e.g. Pannoek, 1910) and at least one visionary work by an autodidact, Mihkel Martna (1860–1934), who used evolutionary theory to frame a utopian dream depicting the future of Estonia (Martna 1900). The reason for the modest contribution of socialists could be that the ideology itself was very much an imported phenomenon (Karjahärm and Sirk 1997, 238). Subsequently an alternative to the two political parties was sought. Such a need was raised in different contexts.

A movement of nationalist intellectuals, calling themselves Noor-Eesti (Young Estonia or New Estonia), was mainly interested in literary, social and political issues but published among other titles translations of popular texts in the natural sciences, e.g. Kliment A. Timiryazev's 1907 book on Darwin,¹⁷ but also one on Haeckel (Jans 1914). Noor-Eesti appeared after the 1905 Revolution. It was critical of political developments dominating the nationalist movement. Its stated goal was to unite socialism with individualism, and to save liberalism from the growing threat of socialism. Finally, the ideals of solidarity were introduced (Ruubel 1918).

Similar concerns were voiced by some members of Ühendus, who blamed the socialists for being too insensitive in national matters and the liberals for forgetting the complex character of society and social issues related to it (Vilms 1915). Ühendus declared the Noor-Eesti as the most promising nationalistic institution representing the elites whose ideas must be followed (H. M. 1926). Radicals too were infected by the ideas of solidarity and many of them chose Ernst Haeckel as their role model. A booklet on Haeckel was issued by this group in 1914 containing a selection of his writings (Jans 1914). It was introduced by Johan Jans (1880–1941), one of the leaders of the Ühendus, with comments favouring monism. For Jans, the Haeckelian approach, even though imperfect, best met current social needs (Jans 1914, 20).

Conclusion

The Estonian reception of Darwin was modest if considered scientifically, and defensive if viewed from the social perspective. By defensive, we mean that the socio-ethnic groups inhabiting present-day Estonia in the second half of the nineteenth and beginning of the twentieth century all perceived the social application of Darwinism as a threat. Social Darwinism in the Estonian context implied the possibility of ethnic extinction, whereby each of the constituent groups was threatened by mighty and aggressive neighbours – for Baltic Germans it was the Russians; for Estonians, it was both Russians and Germans.

The Baltic-German scholars were trying to resist the social-Darwinist model of the advocates of Russification. The latter, besides imposing a struggle for survival, also threatened the feudalized social structure. The Germans' fears were compounded by tendencies emerging from below, by the newly emancipated indigenous ethnic groups – Estonians and Latvians.

¹⁷ Editors' note: on Timiryazev, see Chapter 28 by Eduard Kolchinsky, 'Darwinism and Dialectical Materialism in Soviet Russia', in volume 2.

Despite concerns similar to those of the local Germans – i.e. ethnic extinction – the Estonian reception of Darwinism did also contain positive ideas. The evolutionary model seemed to support national emancipation and to help in breaking down the feudal system. After that had taken place, Darwin's name became quite rare in Estonian public discussions. His works did not appear in Estonian, and the literature was reduced to one biography (Heilborn 1931). (There has been similar neglect concerning K. E. von Baer.) New social concerns brought eugenics and pronatalism to the fore.

In conclusion, we can say that the political situation caused the scientifically minded members of the Baltic-German community to look to alternative scientific theories (biovitalism, monism). The Estonian reception clung finally to authoritarian corporatism, with Francis Galton and Ernst Haeckel contributing more to national ideology than Charles Darwin.